

MSK Pathology DDX Charts

Rev 1.1 · Updated October 2, 2026

Highlighted = Step 1 buzzword

Yellow box = HY · Step 1

1 Bone — metabolic disease

MSK10

Bone disorders — lab values

Disorder	Serum Ca ²⁺	Phosphate	ALP	PTH	Key clue
Osteoporosis	Normal	Normal	Normal	Normal	↓ bone mass, normal mineralization; vertebral compression fractures
Osteopetrosis	Normal (↓ in severe infantile form)	Normal	Normal	Normal	Dense, brittle bone; carbonic anhydrase II ; marrow failure
Paget disease of bone	Normal	Normal	↑↑	Normal	Isolated ↑ ALP ; mosaic bone ; ↑ hat size , hearing loss
Osteomalacia / rickets	↓	↓	↑	↑	Vitamin D deficiency → unmineralized osteoid
1° hyperparathyroidism	↑	↓	↑	↑	Adenoma; osteitis fibrosa cystica , brown tumors , stones
2° hyperparathyroidism (CKD)	↓	↑	↑	↑	Kidney can't excrete phosphate or make calcitriol
Hypervitaminosis D	↑	↑	Normal	↓	Excess vitamin D → abnormal calcification of soft tissues
Sarcoidosis	↑	↑ / normal	Normal	↓	Granuloma macrophages make 1α-hydroxylase → ↑ calcitriol

★ **HY · STEP 1** Isolated ↑ ALP with normal Ca/phosphate = **Paget**. ↑ Ca with ↓ PTH = **PTH-independent** (vitamin D excess, sarcoidosis). Osteoporosis = **all labs normal**.

Metabolic bone disease — what you see

Disorder	Mechanism	Histology / X-ray	Classic clinical clue
Scurvy	Vitamin C deficiency → defective collagen hydroxylation	Defective osteoid; subperiosteal hemorrhage	Bleeding gums , poor wound healing, corkscrew hairs
Osteogenesis imperfecta	COL1A1/COL1A2 → defective type I collagen	Thin, fragile cortex; multiple fractures	Blue sclerae , hearing loss, dentinogenesis imperfecta
Osteomalacia / rickets	↓ Vitamin D → defective mineralization	Wide unmineralized osteoid seams; cupped, frayed metaphyses	Bowed legs, rachitic rosary (children); bone pain (adults)
Hyperparathyroidism	↑ PTH → ↑ osteoclastic resorption	Osteitis fibrosa cystica: tunneling resorption, marrow fibrosis, brown tumors (hemosiderin)	Stones, bones, groans ; ↑ Ca
Osteoporosis	Resorption > formation	Thin, disconnected trabeculae; normal mineral	Vertebral compression and hip fractures in older women
Osteopetrosis	Carbonic anhydrase II → osteoclasts can't acidify	Persistent primary spongiosa, woven bone, no marrow; diffusely sclerotic X-ray (bone-in-bone)	Fractures, pancytopenia , cranial nerve palsies
Paget disease	Lytic → mixed → sclerotic remodeling	Mosaic pattern of cement lines ; thick, disorganized bone	Skull enlargement, hearing loss, bowing; ↑ ALP; osteosarcoma risk

★ **HY · STEP 1** Blue sclerae + multiple fractures in a child = **osteogenesis imperfecta** (type I collagen) — the classic mimic of child abuse. Paget complications: **high-output heart failure** and **osteosarcoma**.

2 Joints and bone infection

MSK11 · MSK23

Joint pain — four categories

Category	Examples	Stiffness / pattern	Joint fluid WBC	Labs / X-ray
Mechanical (non-inflammatory)	Osteoarthritis, injury	Brief "gelling" stiffness; worse with use, better with rest	0 to a few hundred ($\leq 1,000$)	Normal ESR/CRP, RF, ANA; X-ray: joint space narrowing, osteophytes
Inflammatory (autoimmune)	RA, SLE, spondyloarthropathies, PMR	Prolonged morning stiffness ; usually symmetric; fatigue	2,000–50,000 (opaque fluid)	RF, anti-CCP , ANA, $\uparrow$ ESR/CRP
Crystalline	Gout, pseudogout (CPPD)	Acute, intermittent, severe attacks	2,000–50,000 (sometimes higher)	Crystals on arthrocentesis; $\uparrow$ uric acid in gout
Infectious	Bacterial (S aureus), gonococcal, Lyme	Usually monoarticular, hot joint	>75,000	Gram stain / culture

★ **HY · STEP 1** Joint fluid **>75,000 WBC** = treat as septic until proven otherwise. Young, sexually active + monoarthritis/tenosynovitis = **gonococcal**.

Gout vs pseudogout

Feature	Gout	Pseudogout (CPPD)
Crystal	Monosodium urate	Calcium pyrophosphate dihydrate
Shape	Needle-shaped	Rhomboid
Birefringence	Negative (yellow when parallel)	Weakly positive (blue when parallel)
Classic joint	1st MTP (podagra)	Knee
Chronic lesion	Tophi : urate aggregates ringed by foreign-body giant cells (ear, olecranon, Achilles)	Basophilic deposits in cartilage; chondrocalcinosis on X-ray
Associated	Hyperuricemia; Lesch-Nyhan , tumor lysis , thiazide/loop diuretics ; kidney disease	Older age; cartilage degradation; hemochromatosis , hyperparathyroidism

★ **HY · STEP 1** **Needle + negative = gout; rhomboid + positive = pseudogout**. Acute attack $\rightarrow$ colchicine/NSAIDs; allopurinol is for prevention only.

Osteoarthritis vs rheumatoid arthritis

Feature	Osteoarthritis	Rheumatoid arthritis
Mechanism	Mechanical wear of articular cartilage	Autoimmune synovitis $\rightarrow$ pannus
Primary tissue	Articular cartilage	Synovial membrane
Joints	Weight-bearing joints, DIP, PIP, 1st CMC; asymmetric	MCP , PIP, wrists; symmetric; spares DIP
Stiffness / pain	Worse with use, better with rest; brief stiffness	Morning stiffness >1 h ; improves with use
Hand findings	Heberden (DIP) and Bouchard (PIP) nodes = osteophytes	Ulnar deviation , swan-neck , boutonniere ; rheumatoid nodules (soft tissue)
Labs	Normal	RF, anti-CCP (most specific) , $\uparrow$ ESR/CRP
X-ray	Joint space narrowing, osteophytes, subchondral sclerosis and cysts	Erosions, periarticular osteopenia
Gross / histology	Fibrillated cartilage, eburnation , joint mice	Hyperplastic synovium, lymphoid aggregates, pannus
Who	Older age, obesity, prior injury	Women > men; HLA-DR4

★ **HY · STEP 1** DIP involvement $\rightarrow$ think **OA** (or psoriatic), never RA. **Anti-CCP** is the most specific RA test.

Osteomyelitis — acute vs chronic vs tuberculous

Feature	Acute pyogenic	Chronic pyogenic	Tuberculous
Organism	S aureus most common (Salmonella in sickle cell ; Pseudomonas with IV drug use)	S aureus; may be polymicrobial	Mycobacterium tuberculosis (hematogenous from visceral TB)
Site	Children: metaphysis ; adults: epiphysis/subchondral, vertebrae	Same bone, persistent	Spine in ~half (Pott disease)
Histology	Neutrophils, pus, bone necrosis	Sequestrum (dead bone), involucrum (new bone), mononuclear cells, fibrosis	Caseating granulomas ; destructive
Clinical	Fever, localized pain, ↑ ESR	Draining sinus; pathologic fracture; Brodie abscess	Back pain, fever, night sweats; vertebral collapse, psoas abscess

★ **HY · STEP 1** Sickle cell → **Salmonella**; IVDU → **Pseudomonas**; S aureus most common overall. Sequestrum = dead bone; involucrum = new bone around it.

3 Bone tumors

MSK14

Bone tumors — site, age, imaging, histology

Tumor	Typical age	Site in bone	Imaging	Histology / genetics
Osteochondroma	Adolescents (multiple: childhood)	Metaphysis , surface; grows away from joint	Pedunculated bony stalk continuous with marrow	Hyaline cartilage cap ~1–2 cm; EXT1/EXT2
Osteoid osteoma	Young adults (<25), men	Cortex of long bones	Radiolucent nidus <2 cm with sclerotic rim	Woven bone + osteoblasts; night pain relieved by NSAIDs
Osteoblastoma	Young adults	Vertebrae (posterior elements)	Lucent lesion >2 cm	Same histology as osteoid osteoma; pain not relieved by NSAIDs
Enchondroma	Adults 20–50	Medulla of small bones of hands and feet	Well-circumscribed lucency ± ring calcification	Bland hyaline cartilage; multiple = Ollier/Maffucci
Giant cell tumor	Adults 20–40	Epiphysis (extends to subchondral bone), distal femur/proximal tibia	"Soap bubble" eccentric lytic lesion	Osteoclast-like giant cells; stromal cells express RANKL
Aneurysmal bone cyst	Adolescents	Metaphysis, eccentric	Expansile, fluid–fluid levels	Blood-filled spaces with septa of giant cells, spindle cells; USP6
Osteosarcoma	Bimodal: teens; older adults (Paget, radiation)	Metaphysis around the knee	Mixed lytic/blastic; Codman triangle , sunburst	Pleomorphic cells making osteoid ; RB1, TP53
Chondrosarcoma	Adults >45–50	Axial skeleton: pelvis , shoulder, proximal femur	Lytic with ring/arc calcifications, cortical erosion	Atypical chondrocytes permeating bone; IDH1/2; grade predicts behavior
Ewing sarcoma	Children/teens (<20)	Diaphysis ; pelvis	Moth-eaten lytic lesion, "onion skin"	Small round blue cells , CD99* ; t(11;22) EWSR1-FLI1
Metastasis	Older adults	Axial skeleton; often multiple	Lytic (kidney, thyroid) or blastic (prostate)	Carcinoma — most common bone malignancy overall

★ **HY · STEP 1** **Epiphysis** = giant cell tumor (adult). **Metaphysis** = osteosarcoma, osteochondroma. **Diaphysis** = Ewing. Most common benign = osteochondroma; most common primary malignant = osteosarcoma; most common overall = metastasis.

4 Soft tissue tumors

MSK16

Fibrous / myofibroblastic lesions

Lesion	Nature	Who / where	Growth	Histology	Behavior
Nodular fasciitis	Reactive, self-limited (MYH9-USP6)	Young adults (20–40); forearm, often after trauma	Rapid (weeks), 1–3 cm	"Tissue culture" fibroblasts , microcysts, extravasated RBCs	Does not recur
Superficial fibromatosis (Dupuytren)	Benign proliferation	Palm (ring and little fingers), sole, penis	Slow	Bland fibroblasts, dense collagen	Contracture ; may recur
Desmoid (deep fibromatosis)	Locally aggressive (β-catenin/APC)	Shoulder, chest wall, thigh; abdominal wall after pregnancy ; mesentery in Gardner/FAP	Larger, faster	Sweeping fascicles infiltrating muscle	Recur; never metastasizes

★ **HY · STEP 1** Desmoid + colon polyps + osteomas = **Gardner syndrome** (APC). Nodular fasciitis grows fast and looks scary but is **self-limited**.

Fatty tumors

Tumor	Who / where	Key histology	Genetics	Behavior
Lipoma	40–60; subcutaneous back, neck; most common soft tissue tumor	Mature fat, thin capsule	—	Cured by excision
Angiolipoma	Young adults; forearm; often multiple	Fat + capillaries with fibrin thrombi	—	Painful
Spindle / pleomorphic lipoma	Older men; posterior neck / shoulder	Floret giant cells , spindle cells	—	Benign
Well-differentiated liposarcoma	Middle-aged/older; deep thigh, retroperitoneum	Atypical stromal cells, lipoblasts	MDM2 amplification	Recur; can dedifferentiate
Myxoid / round cell liposarcoma	30s–40s; thigh	Lipoblasts, "chicken-wire" capillaries , myxoid stroma; round cells = worse	t(12;16) FUS-DDIT3	Round-cell component = high grade

★ **HY · STEP 1** A deep fatty mass in the **retroperitoneum** = liposarcoma until proven otherwise. **Lipoblast** = the diagnostic cell of liposarcoma.

Rhabdomyosarcoma subtypes

Subtype	Age	Site	Histology	Genetics
Embryonal (most common)	Children <10	Head and neck, genitourinary	Primitive round → spindle "strap" cells	Heterogeneous
Botryoid (embryonal variant)	Young children	Mucosa-lined hollow organs (bladder, vagina)	Grape-like ; subepithelial cambium layer	—
Alveolar	10–25	Deep extremities, head and neck	Clusters of round cells around fibrous septa	t(2;13) or t(1;13) PAX3/7-FOXO1
Pleomorphic	Adults >60, men	Thigh	Large, atypical eosinophilic cells	Heterogeneous

★ **HY · STEP 1** Rhabdomyoblasts: **desmin⁺, myogenin⁺ (MyoD1)**. Grape-like vaginal/bladder mass in a young girl = **sarcoma botryoides**.

Vascular tumors

Tumor	Who / where	Histology	Course / complication
Capillary hemangioma ("strawberry")	Infants and young children; head and neck	Small capillary-sized vessels	Grows rapidly, then involutes
Cavernous hemangioma	Children; head and neck, deeper	Large, dilated, blood-filled vessels	Does not regress; can consume platelets → thrombocytopenia
Angiosarcoma	Older adults (scalp); after radiation (breast); chronic lymphedema (Stewart-Treves)	Atypical endothelial cells forming irregular vessels; higher grade = solid sheets	Aggressive; risks also vinyl chloride, arsenic (liver)

★ **HY · STEP 1** Angiosarcoma of the liver → **vinyl chloride, arsenic, Thorotrast**. Post-mastectomy lymphedema arm mass = **Stewart-Treves**.

Tumor genetics — translocations and amplifications

Tumor	Genetic change
Ewing sarcoma	t(11;22) EWSR1-FLI1
Myxoid / round cell liposarcoma	t(12;16) FUS-DDIT3
Synovial sarcoma	t(X;18) SS18-SSX
Alveolar rhabdomyosarcoma	t(2;13) / t(1;13) PAX3/7-FOXO1
Well-differentiated liposarcoma	MDM2 amplification
Nodular fasciitis	MYH9-USP6
Desmoid	CTNNB1 (β-catenin) or APC
Osteochondroma (hereditary multiple)	EXT1 / EXT2

★ **HY · STEP 1** Translocation ↔ tumor pairs are pure recall on Step 1 — know the four highlighted pairs cold.

Nerve sheath tumors

Tumor	Cells	Relation to nerve	Histology	Syndrome / risk
Neurofibroma	Schwann cells, fibroblasts, perineurial cells	Axons run through the tumor	Wavy "shredded carrot" spindle cells in loose stroma	NF1 (plexiform = pathognomonic ; ~10% MPNST risk)
Schwannoma	Schwann cells only	Eccentric, encapsulated — nerve can be spared	Antoni A (palisading, Verocay bodies) and Antoni B (loose); S-100*	Bilateral vestibular = NF2
MPNST	Malignant Schwann-lineage cells	Arises from a nerve or neurofibroma	Fascicles of atypical spindle cells, necrosis	NF1 (deep plexiform neurofibroma)

★ **HY · STEP 1** **NF1** (chr 17, neurofibromin): café-au-lait, freckling, Lisch nodules, neurofibromas, optic glioma. **NF2** (chr 22, merlin): bilateral vestibular schwannomas.

5 Muscle and nerve

MSK19

Proximal weakness / pain — myopathy vs mimics

Condition	Weakness	Pain & tenderness	ESR	Creatine kinase	Other features
Glucocorticoid-induced myopathy	Proximal (hips first)	No	Normal	Normal	On chronic steroids; type 2 fiber atrophy ; improves when steroids are tapered
Polymyalgia rheumatica	None (pain and stiffness limit movement)	Yes — shoulder and hip girdle , morning stiffness	↑↑ (and CRP)	Normal	Age >50 (usually >60); associated with giant cell arteritis ; dramatic response to low-dose prednisone
Polymyositis	Proximal, symmetric	Mild or none	Normal / ↑	↑ (often up to 50×)	Adults; endomysial CD8+ T cells invading fibers
Dermatomyositis	Proximal, symmetric	Mild or none	Normal / ↑	Normal to ↑	Heliotrope rash, Gottron papules; perifascicular atrophy; malignancy association ; any age (peaks 5–15, 45–65)
Statin-induced myopathy	Proximal	Yes — myalgias common	Normal	↑ (can reach rhabdomyolysis)	Starts after a statin; resolves on stopping

★ **HY · STEP 1** Normal CK + ↑↑ ESR in an older adult with girdle pain = **polymyalgia rheumatica**. Weakness with normal CK and ESR on steroids = **steroid myopathy**.

Polymyositis vs dermatomyositis — histology

Feature	Polymyositis	Dermatomyositis
Age	Adults (>18)	Any age; peaks 5–15 and 45–65
Serum CK	Elevated (often up to 50×)	Normal to elevated
Inflammation location	Endomysial (around individual fibers)	Perimysial / perivascular
Cells	CD8⁺ cytotoxic T cells invading fibers	B cells and CD4⁺ T cells
Atrophy pattern	Scattered necrosis and regeneration	Perifascicular atrophy
Pathogenesis	T-cell-mediated	Antibody/complement-mediated vasculopathy
Extras	—	Skin rash; underlying malignancy

★ **HY · STEP 1** Perifascicular atrophy = dermatomyositis. Both: anti-Jo-1 (anti-synthetase), ↑ CK — screen dermatomyositis for occult cancer.

Muscular dystrophies

Feature	Duchenne	Becker	Myotonic (DM1)
Inheritance	X-linked recessive	X-linked recessive	Autosomal dominant (anticipation)
Defect	Frameshift /nonsense in dystrophin → absent dystrophin	In-frame deletion → reduced, partly functional dystrophin	CTG repeat expansion in DMPK
Onset / course	Early childhood; wheelchair by teens; death ~20s	Later, milder	Adolescence/adulthood (congenital form severe)
Key findings	Gowers sign , calf pseudohypertrophy , ↑↑ CK, dilated cardiomyopathy	Same features, milder	Myotonia (can't release grip), distal weakness, cataracts , frontal balding , testicular atrophy, arrhythmia
Dystrophin stain	Absent	Reduced / patchy	Normal

★ **HY · STEP 1** Frameshift = **Duchenne** (worse); in-frame = **Becker** (milder). Cause of death in DMD: **dilated cardiomyopathy** / respiratory failure.

Myasthenia gravis vs Lambert-Eaton

Feature	Myasthenia gravis	Lambert-Eaton
Antibody target	Postsynaptic nicotinic ACh receptor	Presynaptic voltage-gated Ca²⁺ channel
Weakness	Ocular/bulbar first (ptosis , diplopia); worsens with use	Proximal limbs; improves with use
Reflexes / autonomic	Normal reflexes	Hyporeflexia ; autonomic symptoms (dry mouth)
Association	Thymoma , thymic hyperplasia	Small cell lung cancer
AChE inhibitor	Reverses symptoms	Minimal effect
Diagnosis	Antibodies + EMG; muscle biopsy not needed	Antibodies + EMG; muscle biopsy not needed

★ **HY · STEP 1** MG: **worse with use + thymoma** (CT chest). LEMS: **better with use + small cell lung cancer** (paraneoplastic).

Neuropathy patterns

Feature	Axonal degeneration	Segmental demyelination
Primary target	Axon (neuron)	Schwann cell / myelin
Nerve conduction	↓ amplitude , near-normal velocity	Slowed velocity , conduction block
Biopsy	Degenerating axons, myelin ovoids; fiber loss	Thinly myelinated or bare axons; onion bulbs if chronic
Examples	Diabetes , toxins/drugs, vasculitis, trauma	Guillain-Barré (acute), CIDP (chronic), CMT1
Recovery	Slow (axon regrowth)	Faster (remyelination)

★ **HY · STEP 1** Guillain-Barré: **ascending weakness after infection** (Campylobacter), **areflexia**, **albuminocytologic dissociation** in CSF; macrophages strip myelin.

Shared presentation → distractor vs correct answer

Shared presentation	Common distractor	Correct answer	Deciding clue
Elderly patient with bone pain and ↑ ALP	Bone metastasis	✓ Osteitis deformans (Paget disease)	Mosaic bone pattern, normal $\text{Ca}^{2+}/\text{PO}_4^{3-}$, risk of hearing loss
Elderly patient with brittle bones	Osteitis deformans	✓ Osteopetrosis	Dense, brittle bones; defective osteoclast resorption
Child with growth failure and rickets	Vitamin D deficiency	✓ Fanconi syndrome (proximal tubulopathy)	Multiple proximal tubule defects (acidosis, hypokalemia, hypovolemia from polyuria)
Sickle cell disease with fever and focal bone pain (osteomyelitis)	S aureus osteomyelitis	✓ Salmonella osteomyelitis	Salmonella is most common in sickle cell disease; S aureus most common overall
Elderly patient with joint pain and stiffness	Rheumatoid arthritis	✓ Osteoarthritis	Asymmetric, DIP ± PIP joints, improves with rest
Swollen, painful joint (acute monoarthritis) ± fever	Gout	✓ Septic arthritis	High fever, ↑ WBCs, positive Gram stain from joint aspiration
History of gout (acute attack)	Allopurinol	✓ Colchicine / NSAID	Used for acute attacks; allopurinol is only for prevention
Teen with a painful bone mass	Ewing sarcoma	✓ Osteosarcoma	Metaphyseal involvement, sunburst appearance on X-ray
Child with café-au-lait spots + tumors ± learning disabilities	NF2	✓ NF1	NF1 = neurofibromas, optic gliomas, Lisch nodules; NF2 = bilateral vestibular schwannomas
Ptosis / diplopia	Oculomotor palsy	✓ Myasthenia gravis	AChR antibodies; thymoma; weakness worsens with activity
Proximal weakness	Myasthenia gravis	✓ Lambert-Eaton syndrome	Caudal-to-cranial progression; ↓ presynaptic ACh release, improves with use; small cell lung cancer
Proximal muscle weakness + ↑ CK	Polymyositis	✓ Dermatomyositis	Skin findings (heliotrope rash, Gottron papules)
Acute-onset paralysis	Stroke	✓ Guillain-Barré syndrome	Symmetric, progressive; areflexia and albuminocytologic dissociation in CSF
Painful vesicular rash	HSV	✓ Herpes zoster	Dermatomal distribution, does not cross the midline